



State of the Services

FY 2025 Q1 Report

October 2024 - December 2024

Engagement Events Highlights

Extended Continental Shelf (ECS) Meeting

Washington, DC | December 4-5

- Members of NCEI attended an ECS Task Force meeting hosted by the U.S. State Department to provide development updates since the announcement of the ECS outer limits in December 2023.
- Following a workshop focused on the operational impacts, domestic implementation, and future planning for the project, the State Department held a reception attended by members who helped determine nearly 386,000 square miles of ECS for the United States. During the reception six NOAA NCEI members received letters signed by the Secretary of State for their dedicated efforts on the Task Force.



Industry Proving Grounds Engagement Continues

Virtual | December 10

- The Inflation Reduction Act (IRA) / Industry Proving Grounds (IPG) is supporting small teams of industry professionals to collaboratively identify common user needs within targeted economic sectors and provide input to NCEI's IPG product development.
- NCEI hosted the second IPG meeting with the insurance and reinsurance industry team in December. Furthermore, NCEI has been preparing additional members of the team in partnership with the Reinsurance Association of America to ensure that the team is representative of the sector.

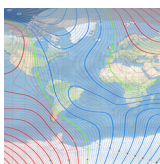


Product and Services Highlights



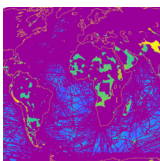
NCEI's Hurricane Helene Response and Recovery

NCEI headquarters in Asheville, NC, was severely impacted by Hurricane Helene on September 27, 2024. The building lost power, water, and communications, disrupting data feeds and essential products. All personnel were safe, and efforts to secure NCEI's data holdings were successful. By December 11, operational ingest of archival data streams was fully restored, and 92% of the ingest backlog was recovered. Data loss was mitigated with support from NCEI's Boulder, CO location. NCEI archives over 229 terabytes of data monthly, supporting research, long-term environmental monitoring, and decision-making across all major economic sectors.



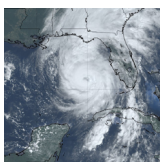
World Magnetic Model 2025

NCEI released the World Magnetic Model (WMM) 2025, including data from both the European Space Agency Swarm tri-satellite constellation and the INTERMAGNET global observatory network. This year's release includes the first ever higher spatial resolution version; the WMM High Resolution. The WMM is a global spherical harmonic model of the Earth's main magnetic field and its changes over time. The WMM is the standard model used by the U.S. Department of Defense (DOD) and multiple international organizations for navigation, attitude, and heading referencing systems using the geomagnetic field, as well as in civilian navigation and heading systems-- such as smartphones.



2025 High Definition Geomagnetic Model (HDGM) and HDGM Real Time

Two new magnetic field Earth models were released by NCEI, the HDGM-2025 and HDGM-2025 Real Time. The HDGM is a global, high resolution model providing values of the main and crustal magnetic fields at any point near Earth's surface. The HDGM Real Time additionally provides magnetic fields generated by electric currents in the magnetosphere and ionosphere. The new models include a new crustal field derived from an updated crustal anomaly grid, a new forecasting algorithm for the Disturbance Storm-Time index, and updated main field and secular variation models until December 2025.



2024 U.S. Billion-Dollar Weather and Climate Disasters

In 2024, the U.S. experienced 27 billion-dollar weather and climate disasters. This is the second-highest disaster count on record. These disasters included: 17 severe storms, five tropical cyclones, two winter storms, one flooding event, one drought/heat wave, and one wildfire event.

Stakeholder Highlights



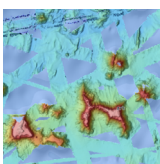
NCEI participated in the Space Weather Scales Workshop and Space Weather as a Global Challenge event, which were held in support of the U.S. Implementation Plan of the National Space Weather Strategy and Action Plan released in 2023. The events offered the opportunity to advance the goals outlined in the Action Plan, particularly in the area of international coordination, cooperation, and engagement when responding to extreme space weather events. Following the event a small meeting between international participants and U.S. government Space Weather leaders was held at the White House Office of Science and Technology Policy to discuss current and future opportunities to partner on space weather topics, including observation sharing, modeling, and resilience.



As part of the Regional Climate Center (RCC) 40th anniversary celebrations in 2023, the six RCCs collaborated on a Bulletin of the American Meteorology Society article which was released in the October 2024 issue. The article highlights the history of the RCC program, the evolution of climate services, RCC services in action, and the future of regional climate services.



NCEI staff delivered a webinar at the Low Earth Orbit Satellite Science Seminar to 85 participants in December. The webinar provided a summary of the September 2024 Alaska Next Generation Satellite Workshop to National Environmental Satellite, Data, and Information Service (NESDIS) leadership and other core partners. User needs were identified and recommendations drawn from attendees and speakers.



NCEI representatives recently wrapped up Phase 2 of the DOD Naval Maritime Working Group effort to design and establish a global Bathymetric Data Repository (BDR). The team brought together subject matter experts, data owners, and data users to design and establish requirements for a global BDR. The BDR will allow DOD users to quickly and easily access bathymetric and shoreline data in areas of operation and will be crucial for mission planning, execution, and resource allocation.

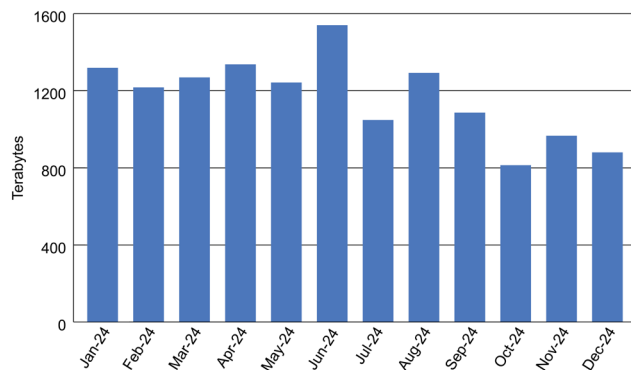
Upcoming Engagements

American Meteorological Society Annual Meeting	Jan. 12-16
Joint Climate Prediction Applications Science Workshop and Climate Diagnostics & Prediction Workshop Meeting	Mar. 24-27
NESDIS User Engagement Summit	May 5-7

Customer Reach

NCEI Makes Data Accessible

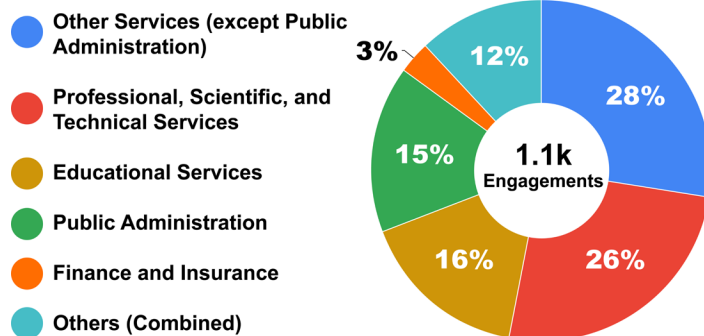
- 827M web hits and 2.66k TB of data delivered online*



*Analytics include model & satellite data; data statistics may not be comprehensive across NCEI (e.g. CLASS)

U.S. Sectors Served

- Top Sectors



Unique Customer Requests

NCEI Data Informs Dive Sites on Lake Michigan:

A recreational dive team contacted NCEI regarding the wreck of *Lady Elgin*, a 250 ft paddle steamship that sank and broke apart in 1860, about nine miles offshore of Lake Michigan. NCEI provided the team with 2023 NOAA survey data, leading to their discovery of a previously unknown debris field containing objects 1-5 meters in diameter, stretching from the skeleton frame of the sunken vessel to the boilers. They plan to dive these features next year to investigate further.

Disney Parks & Resorts Data Request:

The Walt Disney Company Safety Team contacted NCEI for worldwide temperature, humidity, heat index, and wet bulb globe temperature data for locations near Disney Parks & Resorts. NCEI provided Applications Program Interface access to historical hourly data from the Local Climatological Data, which will assist the Disney Team in satisfying federal health and safety requirements for their operations.

Fans



132.6k



64.6k



22.9k

